

4 A

Forces on all 3 masses:

$$F_{\text{floor on Z}} - (m_X + m_Y + m_Z)g = (m_X + m_Y + m_Z)a$$

$$F_{\text{floor on Z}} - 9mg = 9ma$$

$$F_{\text{floor on Z}} = 9mg + 9ma$$

Forces on Z alone:

$$F_{\text{floor on Z}} - F_{Y \text{ on } Z} - m_Z g = m_Z a$$

$$F_{\text{floor on Z}} - F_{Y \text{ on } Z} - m_Z g = m_Z a$$

$$F_{Y \text{ on } Z} = F_{\text{floor on Z}} - m_Z g - m_Z a$$

$$F_{Y \text{ on } Z} = 9mg + 9ma - 5mg - 5ma = 4mg + 4ma$$

